

0 2 . 1

Describe how a phosphodiester bond is formed between two nucleotides within a DNA molecule.

[2 marks]

0 2 . 2

The two DNA strands of a particular gene contain 168 guanine bases between them. The relationship between the numbers of guanine bases (G), adenine bases (A), thymine bases (T) and cytosine bases (C) in these two strands of DNA is shown in the following equation.

$$G = 4(A + T) - C$$

Use this information and your understanding of DNA structure to calculate the maximum number of amino acids coded by this gene.

Show your working.

[2 marks]

Answer _____

0 2 . 3

Name the protein associated with DNA in a chromosome.

[1 mark]



0 2 . 4

In the process of semi-conservative DNA replication, the two strands within a DNA molecule are separated. Each then acts as a template for the formation of a new complementary strand.

Describe how the separation of strands occurs.

[2 marks]

7

Turn over for the next question

Turn over ►

